Structured Generation, XTML, and Walle

Deep Dive into Kimi K3 — Chapter 27

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Foundation Books Project

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Goal: follow one release report, `studio-contract-001`, through schema admission, Walle preparation, legal-token decoding, XTML structure, and call/result identity, and name exactly which predicate each check establishes.

Roadmap

Values, schemas, and preparation

Legal next tokens

XTML structure and call identity

The validity ladder and its boundary

Values, schemas, and preparation

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- MFJS is a deliberately small schema language; a support claim is a triple (feature, level, revision) at Walle rev. `e5c4ecf3`.
- Walle canonicalization returns an atomic pair (S', d). Byte stability is not equivalence: dropping `maximum 12` is stable yet admits 13. [OBSERVED: 14 public-Walle pairs: 9 equal on tested domains, 5 weakened]

Legal next tokens

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- Sound, prefix-complete, termination-safe decoding proves structural membership only: `ready` and `not_ready` are both in $L(G)$.

XTML structure and call identity

Carrier, channel, scope, and identity

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  [REPORTED: K3 App. F]

Carrier, channel, scope, and identity

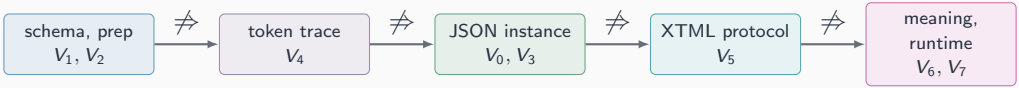
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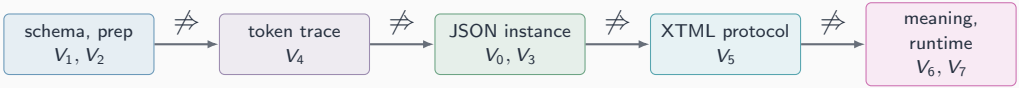
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- Zones and option scopes give one-shot options a one-message lifetime; only a tools-channel call registers a pending call.
- Parallel results arriving b, a resolve by opaque ID within block B_A before any normalized view shows a, b — never by arrival rank.

The validity ladder and its boundary

Eight predicates, one honest handoff

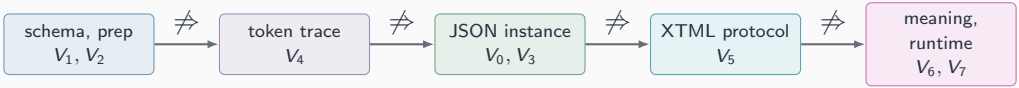


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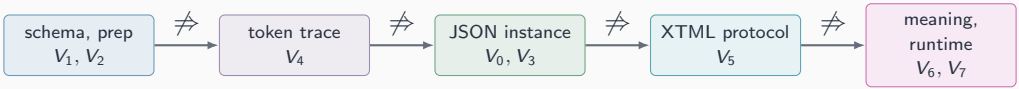
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Structural recovery stops before authority. Next: Chapter 28's Tool Execution, Agent State, and Durable Context.